

Water Industry Pack Guide

The Industry Pack is a collection of resources for use in the Water/Wastewater industry. Starting with the simple tanks, pumps and valves from the Water Treatment Demo we also included advanced components such as VFD and peristaltic pumps. Each component is paired with a UDT so you can quickly link up live PLC tags and visualize your data. Components are themable and include appearance selections and optional displays such as sparkline charts. Since everything is built in Ignition, you can customize everything.

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INSTALLATION

1. Import project file. You may import as a new project from the Gateway Webpage or import select resources from the Ignition Designer.

2. Import tags.

- a. ****Required****. Import udt.json into `_UDT Definition_` first.
- b. ***Optional***. Import tags.json into `_Tags_` after importing the UDTs. This is only necessary for viewing the example tags and demo resources view.

3. Create OPC UA Device Connection

- Select the Programmable Device Simulator.
- Name the newly created device simulator water-pack-sim.
- Select More > Edit Program for the device simulator.
- Load the water-pack-instructions.csv into the simulator.

SETUP

UDT

Choose a UDT matching your equipment. You may want to customize the included UDTs or create a new UDT using the provided types as a parent. Note that customizing the UDTs may break the demo tags but these tags are provided only for examples are not required for any of the resources to work correctly.

Parameters

Start with the parameters. We have included a name and units parameter. The name will be used in equipment displays. The units value is copied over to the engineering units for any OPC value in the UDT. Additional parameters may be needed for OPC addressing or other customizations.

OPC Values

Next take note of OPC tags. These are tags that will require tag addressing. Here you may add new values for your equipment or remove existing tags. Use care when removing tags as this may break the components.

Setpoints

Each tag values has a high and low value in the Setpoints folder. The analog indicator and text-state styles rely on these setpoints. These values must be populated for color coding and analog indicators to work. When adding new values this is where you will want to put high and low setpoints. You may also configure alarms based on the setpoints.

History and Alarming

Set up any desired history and alarming. The history will be used for sparklines and other trends.

TAGS

Start by creating at least one instance of a included UDT with addressing.

DISPLAY

Choose the component corresponding to the tag in the components folder. Use these as embedded views and refer to the documentation to fill in the parameters. Most resources require a path to the tag.

DEMO

All installation steps are required in order to view the demo. All project files, UDTs, tags, and the programmable device simulator must be installed. After installation, open a designer and open the Main view. This view provides an samples overview using all the resources, and a list of resources.

The resources are listed along the left hand side. Documentation is provided with each resource. A sample view is displayed on the Overview selection. This view can be found at Exchange/WaterPack/Demo/Resources

RESOURCES

Analog Indicator

Display for one value with defined setpoints. There is a value display with label and units and an analog indicator. Value and indicator displays are both optional. The resource is linked to one tag within one UDT instance. The tag must have setpoints defined.

Filter

Filter component displays the current status and key values. The filter resource pairs with an instance of a UDT for real time display of the filter state, flow, level, loss of head and turbidity values. Optionally an analog indicator may be displayed for each numerical value.

List of Instances

This resource will list all instances of a given UDT. All tag values at the root level of the UDT path are included.

Tank

Tank graphic with level and optional spark line and scale. Like all resources the tank includes mimic, simple and P&ID versions. In addition, the tank has a special “tank” version with a cylindrical tank. The tank resource also pairs with an UDT.

Valve

Four way valve with optional handle and open percentage. The valve resource pairs with an UDT.

Pumps: Standard, VFD and Peristaltic

Three pump resources provide options for a standard pump, VPD, and a peristaltic pump with more advanced graphics. Each pump resource pairs with an UDT.

Sparkline

Sparkline component with label for number of hours trended. This resource is embedded with the tank resource and can also be used to trend any single value.

Click to Graph

Small icon that can be configured to open a trend view for any single tag. This resource is included with the level value on the tank view. Note using this resource requires setting the [default]Exchange/WaterPack/Meta/TagGatewayName to your tag gateway name.

HOA Control

Clicking a tank, valve, pump or filter component will open an HOA control popup. It is recommended to add elevated security restrictions to this popup.

Setpoint Control

Clicking the gear icon on the HOA Control popup will open the Setpoint Control popup. This popup provides write access to all tags in the UDT Setpoints folder along with the In Service tag. It is recommended to add elevated security restrictions to this popup.

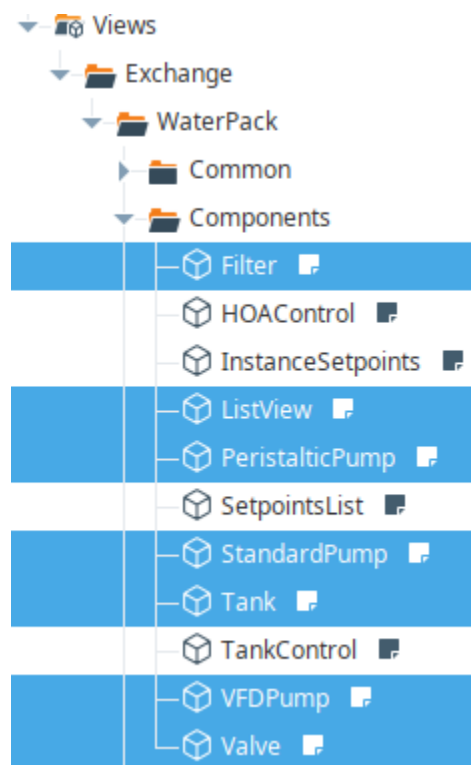
RESOURCE USAGE

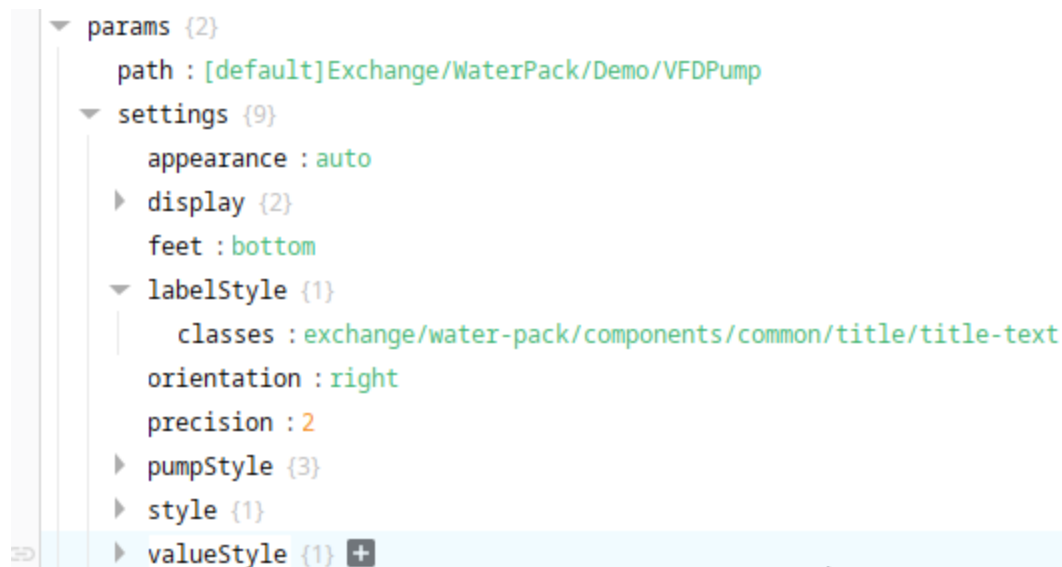
The following resources may be used as embedded views: Filter, List View, Peristaltic Pump, Standard Pump, Tank, VFD Pump and Valve. These resources can be used as part of an HMI diagram on a coordinate container or in any other view such as a dashboard or popup. The list view may be especially useful as a status on a dashboard.

The HOAControl, InstanceSetpoints, SetpointsList and TankControl views are all used as popups and logic is in place to open these views when clicking resources.

Resource Parameters

The Filter, Tank, Valve and Pumps resources all have two main parameters: path and settings. The path parameter is bound to a single UDT instance representing the device. The settings parameter is a collection of settings mostly related to the appearance and styling. Any default appearance and style classes can be overridden using the settings parameters.





USER DEFINED TYPES

The pump, valve and tank resources are linked to a UDT. Using the Industry Pack resources requires UDT setup. It is recommended to create new UDTs inherited from the standard UDTs included with the pack.

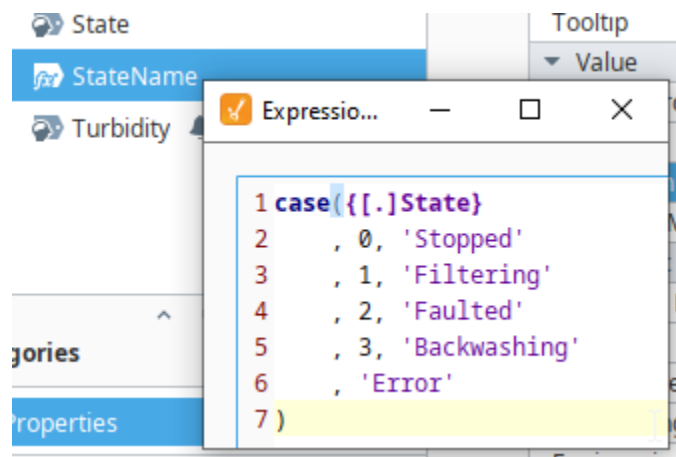
UDT Values

For demo purposes most standard UDT values are memory tags. To use the tags with PLC values, change the Value Source to OPC and set the OPC Server and OPC Item Path.

Loopup	
Value	
Value Source	OPC
Data Type	Double
OPC Server	
OPC Item Path	
Numeric	

UDT States

Each UDT has an integer state value intended to be bound to an integer PLC value for device state. The state text display is determined by the State Name expression tag. Set this tag to customize the text displayed with each state. Note that 0 is reserved for stopped, 1 for on or running and 2 for error or faulted.



UDT Setpoints

Most UDTs include a Setpoints folder meant to hold all set points for the UDT instance. Set point control popups are specially designed to provide write access to tags in this folder. Built in alarms depend on values in this folder. Custom set points should be added here.

Units of Measure

Each UDT has a parameter for units which is then bound to the engineering units for the associated tag. This value is then displayed on screen.

Parameters

Name	Data Ty...	Value
name	String	
path	String	
units	String	

Note in this example the Peristaltic Pump flow engineer unit is bound to the units parameter

Type Structure

- PeristalticPump
 - Setpoints
 - Flow
 - InService
 - Mode
 - ModeName

Categories

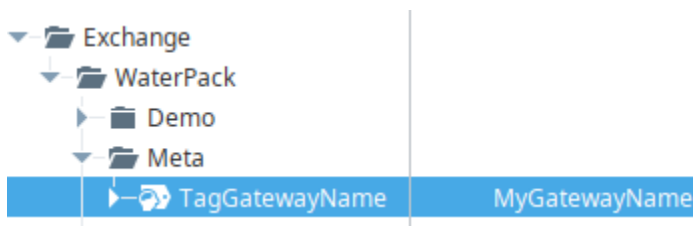
- All Properties
- Basic
- Meta Data
- Value

All Properties

Data Type	Float
Expression	case({[.]Mode}, 0, 0.0 // Off, 1, tag({pat...
Execution Mode	Event Driven
Numeric	
Deadband Mode	Absolute
Deadband	0.0001
Scale Mode	Off
Engineering Units	{units}
Engineering Low Limit	0.0
Engineering High Li...	100.0
Engineering Limit M...	No_Clamp
Format String	#,##0.##
Security	

Tag Gateway

The tag gateway name setting is used to add pens to the power chart when using the click to graph feature.

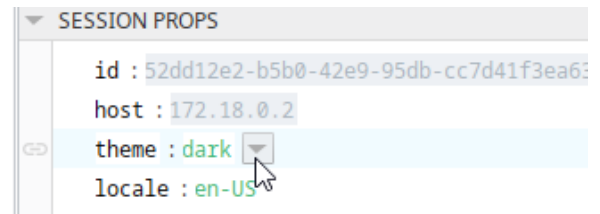


Set this to your Ignition Gateway name if you are using a single gateway. If your tags are on a remote tag provider you will need to provide the remote gateway name.

STYLING

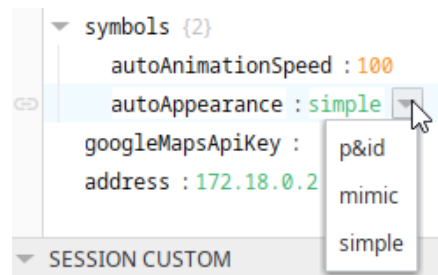
Theme

Perspective theme colors are used in place of hard coded colors. Changing the theme session property will change the project theme.



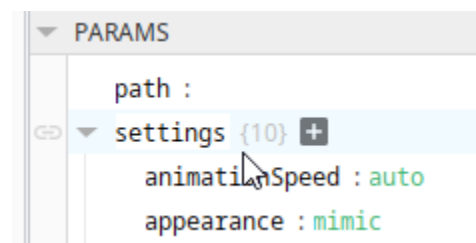
Appearance and Animation

The tank, pump and valve resource are all made to behave like the built in symbol components. Each resource has a P&ID, Mimic and Simple version. By default this will follow the built in symbol autoAppearance session property.



Any resources with animation will use the session property for auto animation speed unless otherwise specified. This speed is a percentage value. By default 100 equates to four seconds animation duration.

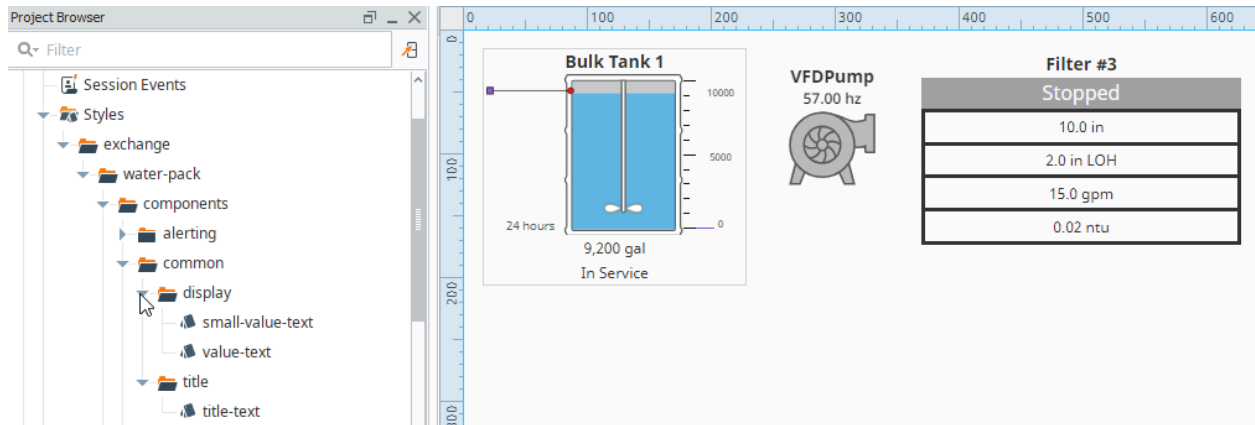
Each resource has an override for appearance and animation under the settings parameter. Leave this setting on auto to use the session property values. Alternatively these can be manually overwritten. In this example we have let the animation speed at the default setting and changed the appearance to mimic.



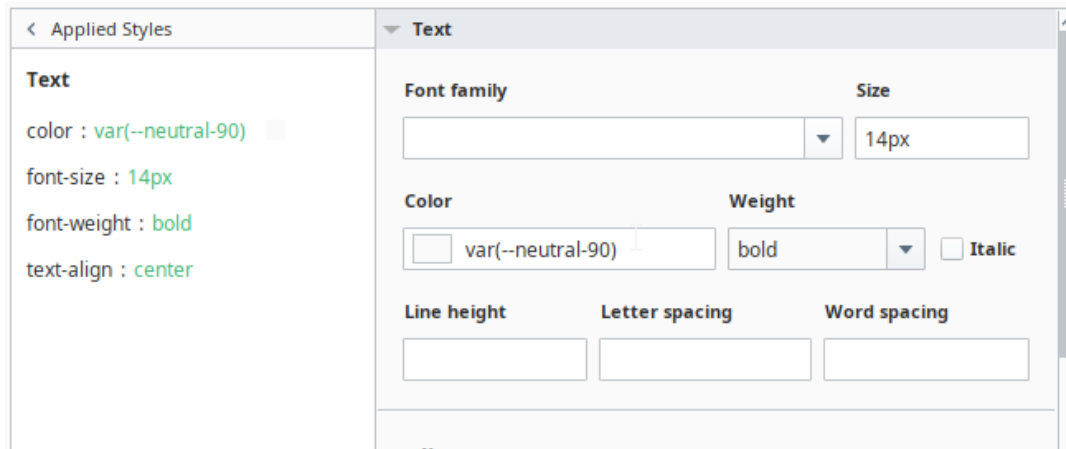
Set animation to zero to turn off animation. Symbol animations only activate when the represented device is in a running state.

Style Classes

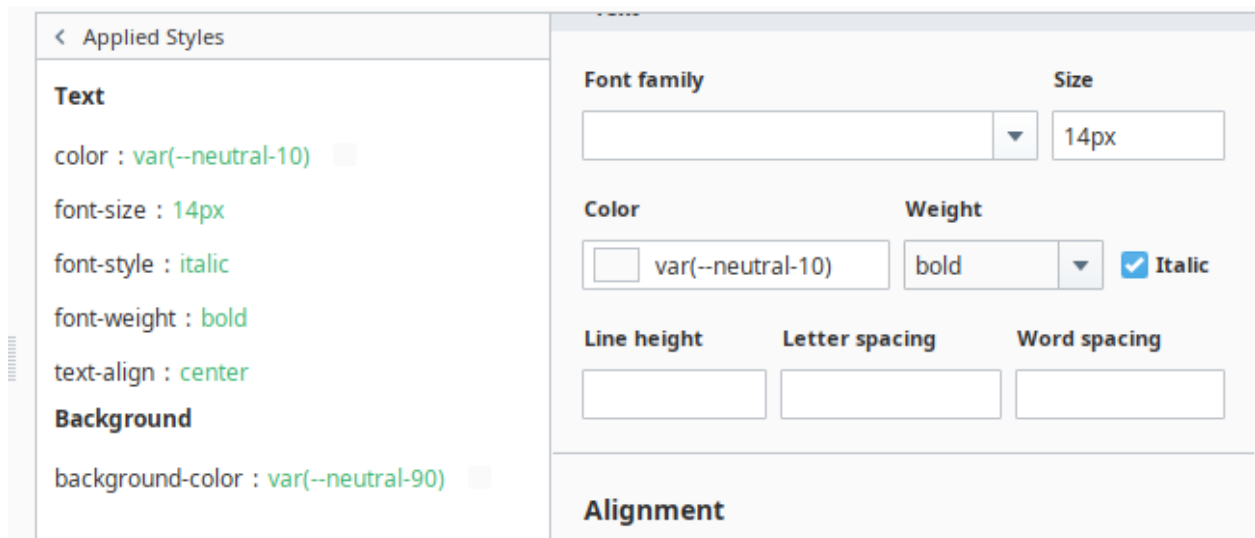
The style classes allow you to change the look and feel of all components in one place. For example the title text style will change the name label on most components.



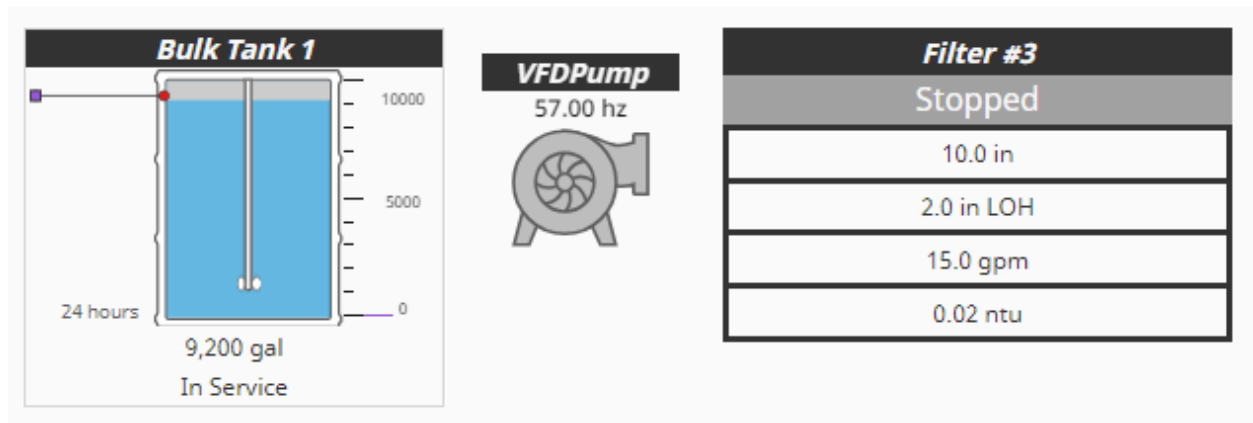
Updating the title text would change the appearance overall giving you a fast and powerful method for branding your project. In the above example we are using the default style for title text. We use the neutral-90 text color and a bold 14 pixel font.



Let's give the titles a background color and contrasting text



The style class update is applied to all components



State Based Style

The symbol-state and text-state style classes are used in conjunction with device state. This allows overall styling for running, stopped and faulted devices.

The pump component is shown below styled with the default running, stopped and faulted states. The pack styles can be customized to fit your standards. For example, changing the symbol-state running class to use a green fill will update all component instances that are in a running state.

